Image Denoising Technique for CT Dose Modulation

Haneen A. Elyamani
Image Science Lab., Dept. of Math., Faculty of Science, Suez Canal University, Ismailia, Egypt
haneen_elyamani@yahoo.com

Samir A. El-Seoud
Faculty of Informatics and Computer Science, The British University in Egypt, Cairo, Egypt
samir.elseoud@bue.edu.eg

Essam A. Rashed
Faculty of Informatics and Computer Science, The British University in Egypt, Cairo, Egypt
essam.rashed@bue.edu.eg

ABSTRACT
Low-dose computed tomography (LDCT) imaging is considerably recommended for use in clinical CT scanning because of growing fears over excessive radiation exposure. Automatic exposure control (AEC) is one of the methods used in dose reduction techniques that have been implemented clinically comprise showing significant decrease scan range. The quality of some images may be roughly degraded with noise and streak artifacts due to x-ray flux, based on modulating radiation dose in the angular and slice directions. In 2005, the nonlocal means (NLM) algorithm showed high performance in denoising images corrupted by LDCT. The proposed method incorporates a prior knowledge obtained from previous high-quality CT slices to improve low-quality CT slice during the filtering process because of the anatomical similarity between the arranged image slices of the scans. The proposed method is evaluated using real data and CT image quality is notably improved.

CCS Concepts
• **Computing methodologies→Image processing.**

Keywords
X-ray CT; denoising; dose reduction; nonlocal means filtering

1. INTRODUCTION
X-ray computed tomography (CT) has been widely used in medical domain for diagnosis, image-guided radiotherapy, screening, and image-guided surgeries. CT scan is a radiation-intensive procedure [1]. However, the high radiation dose delivered to the patients has potential harmful effects including cancerous disease and genetic diseases [2]. This fact raised safety concerns in the medical physics community. Therefore, high-quality CT images reconstructed from low-dose CT are highly desired. Hence, several models have been proposed for dose reduction in CT scan to decrease radiation related risks [3,4].

Recent developments in CT technology, including implementation of automatic exposure control and optimizing the system parameters [5]. Automatic exposure control (also known as automatic tube current modulation (ATCM) or dose modulation) is implemented on CT scanners allow reduction in radiation exposure during a scan length and amongst patients to meet a certain image quality level [6, 7]. ATCM systems determine the patient attenuation and describe changes to the scanner output specially designed to the specific patient and body region to meet the required image quality and especially useful for body parts that are not uniform in size or non-homogenous scan regions as shown in Fig. 1. On the other hand, dose reduction by lowering the tube current lead to a noisy image and artifacts, which effect of the diagnostic accuracy. Therefore, there is a need to improve a technique which could denoise images in ATCM without losing image quality.

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ICSIE '18, May 2–4, 2018, Cairo, Egypt
© 2018 Association for Computing Machinery.
ACM ISBN 978-1-4503-6469-0/18/05…$15.00
DOI:https://doi.org/10.1145/3220267.3220288

Non-local means (NLM) denoising algorithm is an effective image denoising technique that exploits the high degree of redundant information present in most images. It uses the self-similarity property of images to suppress noise by replacing the intensity of each pixel with a weighted average of its neighbors according to similarity based on patches.

In this paper, we propose a novel approach for signal to noise ratio (SNR) enhancement of real chest images based on the NLM algorithm by using series of images with different level of noises result by using ATCM. We consider the case where the ATCM is not accurate such that the dose is extremely low in some slices which causes significant artifacts. However, some neighbor CT slices can be found with relatively high-dose that result in high-quality images. The information extracted from high-quality slices are used as a searching domain for a NLM-based algorithm to improve the quality of low-quality slices. Evaluating by real data exhibits that the proposed method leads to the improvement of image quality.

The remainder of this paper is organized as follows: Section 2 is an overview of the NLM algorithm. In Section 3, we describe the proposed method for tube current modulation. Section 4 describe the experimental results and Section 5 discusses the results and present conclusions.

2. OVERVIEW OF NLM ALGORITHM
Buades *et al.* [8] proposed the classical formulation to the NLM algorithm. It assumes that the images have similarity of elements and patterns. When filtering each pixel, the filter searches for similar pixels from throughout the local image, named search window (SW), giving equal weight to neighboring and non-neighboring local image regions.

Due to the success of the NLM filter, many of regularization models has been based to NLM filter were also proposed for different inverse problems in image denoising. One of regularization model uses previous high-quality scan (preHQ) image instead of using noisy image based on the presence of a prior radiography of the patient [9]. Some patients need to repeat CT scans that in several clinical applications such as longitudinal

studies, disease monitoring, and image-guided radiotherapy. To improve the benefit of radiation dose, the initial scan can be acquired with high quality to set up a guide, followed by a series of low-dose scans.

A preHQ-guided NLM filtering method proposed by Ma *et al.* [10] for CT image. Assume that $x = (x_1, x_2, \dots, x_n)$ is the pixel in an image, the NLM corresponding to image pixel x_i^{LD} (noisy image) is a weighted sum of arbitrary image pixels $x_j^{preHQ\ registered}$ (reference image) that satisfy some statistical properties.

$$guidedNLM(x_i^{LD}) = \sum_{j \in SW_j} \omega_{ij}(x_i^{LD}, x_j^{preHQ\ registered})\, x_j^{preHQ\ registered} \quad (1)$$

where x_i^{LD}represents the pixels in noisy image, $x_j^{preHQ\ registered}$ represents the pixels in previous high quality image that recorded for the current low-dose image, $\omega_{ij}(x_i^{LD}, x_j^{preHQ\ registered})$ is the weighting coefficient and $guidedNLM(x_i^{LD})$ is the new intensity value of the pixel located at x_i^{LD} after filtering, the weighting coefficient satisfy the constrains $0 \le \omega_{ij}\left(x_i^{LD}, x_j^{preHQ\ registered}\right) \le 1$ and $\sum_{j \in SW_j} \omega_{ij}(x_i^{LD}, x_j^{preHQ\ registered}) = 1$ and was given as:

$$\omega_{ij}\left(x_i^{LD}, x_j^{preHQ\ registered}\right) = \frac{exp\left(-\frac{\left\|V_i^{LD} - V_j^{preHQ\ registered}\right\|_2^2}{h^2}\right)}{\sum_{j \in SW_j} exp\left(-\frac{\left\|V_i^{LD} - V_j^{preHQ\ registered}\right\|_2^2}{h^2}\right)} \quad (2)$$

A V_i^{LD} and $V_j^{preHQ\ registered}$ are two patches with the same size that centered at pixels x_i^{LD} and $x_j^{preHQ\ registered}$ (called patch-window (PW), e.g. 5×5 pixels in 2D case), h is the filtering parameter that controls and $\|.\|_2$ is the Euclidean distance between two patch windows in high dimensional space.

3. PROPOSED METHOD

The preHQ guided filtering results may be further improved rather NLM algorithm. However, previous high-quality image of the same patient is not always available. Using ATCM techniques in CT scanners, where using different x-ray doses may result in different quality images. The goal here is to use the neighbor high-quality slices to improve the successor low-quality slices within the framework of NLM. Authors developed an adaptive approach using NLM for CT image denoise in [11]. In this method, the searching window is limited to pixels belongs to the same slice. In the current work, this method is extended for CT dose modulation applications. The proposed method can be implemented as follows:

[**STEP 1**] Acquire few slices ($x^{(1)}$ *to* $x^{(t)}$) with high-dose CT.

[**STEP 2**] Gradually reduce the x-ray power along with patient bed motion for slices ($x^{(t+1)} : x^{(n)}$).

[**STEP 3**] For each $x^{(i)}$, $t<i<=n$, use the NLM with search window $x^{(j)}$, $i\text{-}s<j<i$. *i.e.* use the preceeding *s* slices as search domain.

[**STEP 4**] Repeat [STEP 3] until stopping criteria is satisfied.

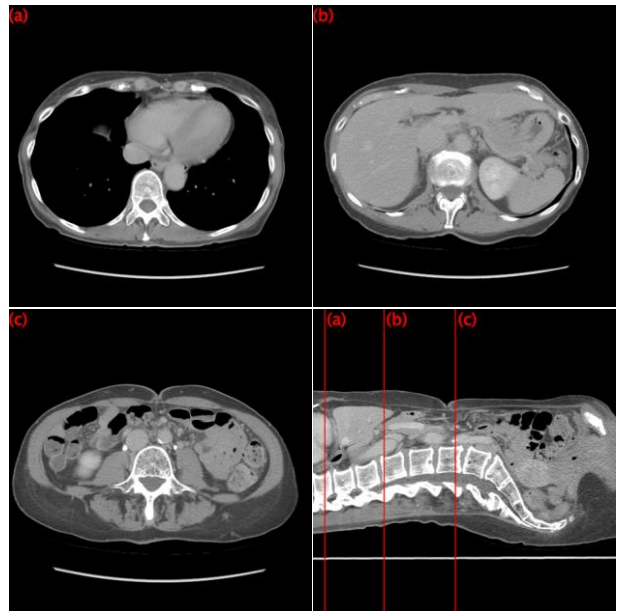

Figure 1: (a)-(c) are different slices representing different tissue densities that requires CT dose modulation. Corresponding position in sagittal pan view is shown in bottom right image.

The proposed method is called ATCM-NLM. The initial high-dose slices are used as a search window for NLM to improve the next slice. However, this process can be used for only few slices with local neighborhood of high-dose slices. The improved slices are then used as search widow to later ones, and so on. It is expected that the image quality improvement with gradually decrease the statistical noise problem. However, the efficiency of the NLM filtering is becoming weaker when the filtered slice is of far distance from the high-dose slice. Therefore, this process is repeated several times (STEP 4), until a reasonable image quality is achieved.

4. EXPERIMENTAL RESULTS

4.1 Image Quality Measures

To evaluate the performance and estimate the image quality of the proposed method, we used the image quality measures of the peak signal-to-noise ratio (PSNR).

$$PSNR(x) = 20.\log_{10}\left(\frac{MAX(x)}{\sqrt{MSE(x)}}\right) \quad (3)$$

where MSE is the mean square error and is computed as follows

$$MSE(x) = \frac{1}{n}\sum_{i=1}^{n}(x_i - x_i^*)^2 \quad (4)$$

where $x^* = (x_1^*, x_2^*, \dots, x_i^*)$ is the pixel value of true reference image and $x = (x_1, x_2, \dots, x_n)$ is the corresponding denoised image. Also, relative root mean square error (RRME).

$$RRME(x) = \sqrt{\frac{\sum_{i=1}^{n}(x_i - x_i^*)^2}{\sum_{i=1}^{n}(x_i^*)^2}} \quad (5)$$

Moreover, evaluations using RRME and PSNR measurements are

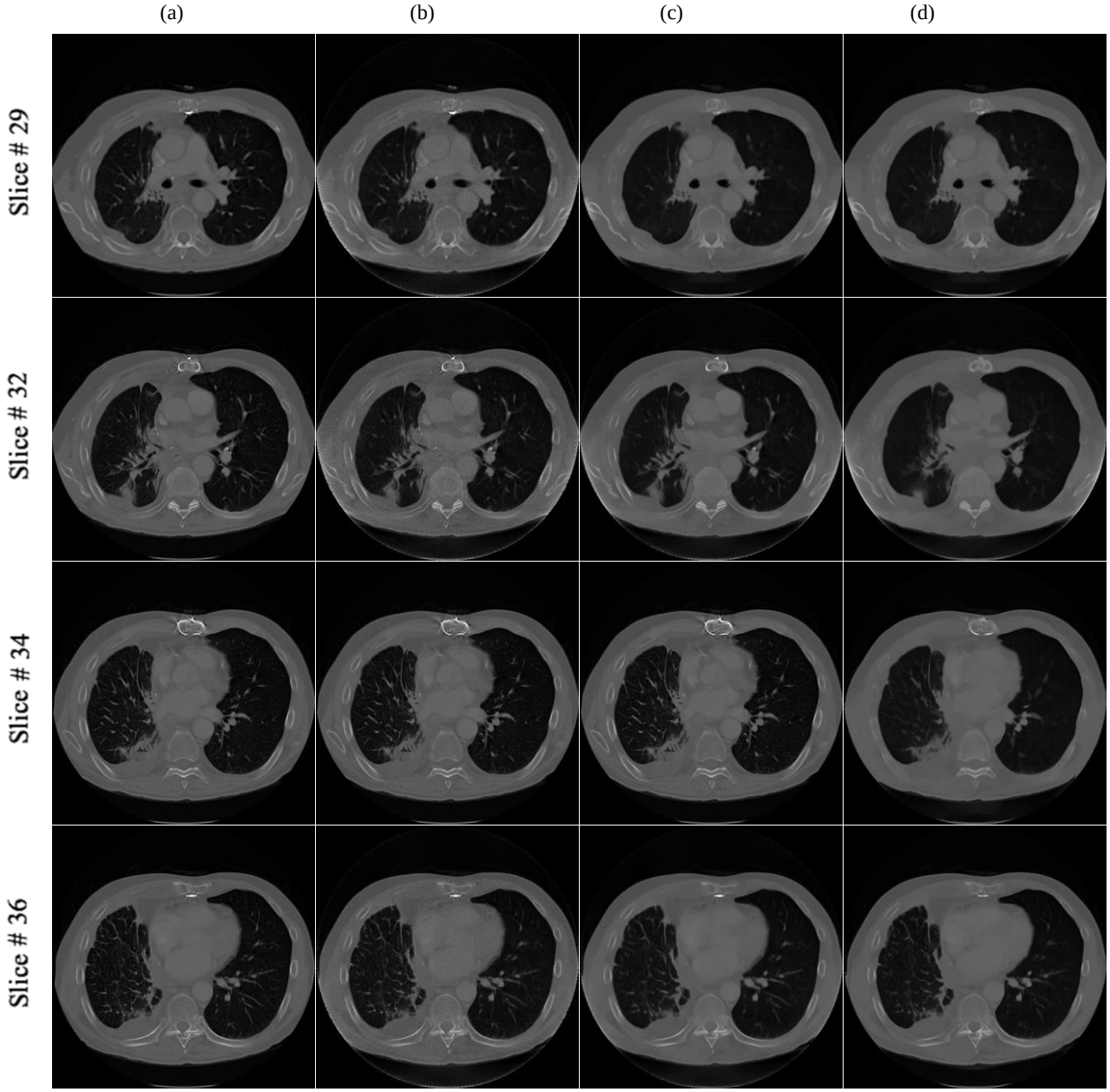

Fig. 2: Result for different slices (a) True image, (b) noisy image, (c) denoised image using standard NLM algorithm [8], and (d) denoised image using proposed ATCM-NLM algorithm.

4.2 Results

We have used a CT volume representing chest imaging. An artificial statistical noise is added within projection domain and image is reconstructed to simulate low-dose CT. Figure 2(a) shows the images from the normal dose scan and the images with low-dose noise are shown in Fig. 2 (b). Images restored from the low dose scan by using the standard NLM algorithm and the proposed ATCM-NLM algorithm are shown in Fig 2 (c) and (d), respectively. Fig. 3 shows the zoomed images corresponding to Figure 1. It can be observed that noise in the low-dose CT images is effectively suppressed using the ATCM-NLM method. used to validate the effect of ATCM-NLM method and results are illustrated in Fig. 4. It is also observed that the proposed method has a high effect on the final image quality. The reason for this image quality improvement is explained as follows. When the search domain for arbitrary pixel is set to the same low-quality slice, it is likely that the selected set of pixels are suffering from noise and other artifacts. One the other hand, when the search domain is set to a high-quality neighbor slice, it would likely lead to higher quality as the anatomical structures are not changed significantly and the image pixels are not suffering from low-dose noise. This proposed method can be easily implemented and can

lead to produce a diagnosis useful image of low-dose for the sake of patient protection.

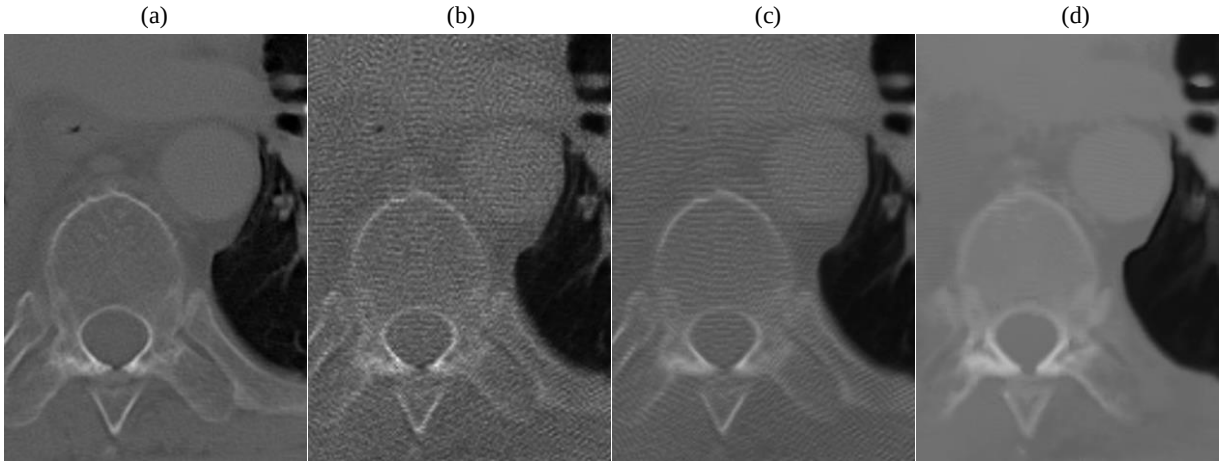

Figure 3: Zoomed ROIs corresponding to Figure 2 slice 34: (a) the normal dose scan (b) the low dose scan (c) the restored image from the low-dose scan by NLM algorithm (d) the restored image from the low-dose scan by the ATCM-NLM method.

5. CONCLUSION

This paper presents ATCM-NLM method has a strong power in detecting noise and image restoration may already be able to generate well image quality for LDCT because filtering in current low dose image is usually problematic, the introduction of high quality images can significantly enhance image. Experimental results show that the proposed method leads to a notable enhancement in image quality.

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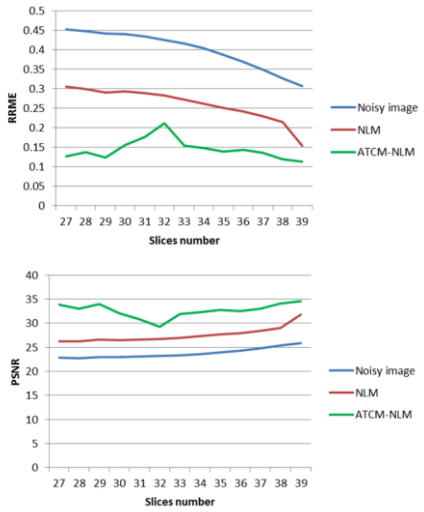

Figure 4. RRME and PSNR with different of slices. Top is RRME curves and bottom is PSNR curves.

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